

TAKE-HOME QUIZ 3

MATH 32A

Note: Due Feb 21 Friday at 11:59pm

There are three questions in total.

1. Evaluate the following limit or determine that it does not exist

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{xy^2}$$

2. Evaluate the following limit or determine that it does not exist

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2|y|}{x^2 + y^2}$$

3. Compute the following:

(a) $\frac{\partial}{\partial x} \left(\frac{y}{x+y} \right)$

(b) $\frac{\partial}{\partial u} \ln(u^2 + uv)$

(c) $f_{yy}(2, 3)$, where $f(x, y) = x \ln(y^2)$